

TrES-2b

R-1630

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_150710 · created 2026-08-02T05:39:23+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457213.90436 +/- 0.00037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.038 +/- 0.042
Transit depth (Rp/Rs)^2	0.15 +/- 0.32 [%]
Orbital Inclination (inc)	84.27 +/- 2.85
Airmass coefficient 1 (a1)	0.69 +/- 0.14
Airmass coefficient 2 (a2)	0.3 +/- 0.27
Scatter in the residuals of the lightcurve fit is	19.72 %
Best Comparison Star	#1 - [221, 348]
Optimal Aperture	5.91
Optimal Annulus	10.84
Transit Duration (day)	0.055 +/- 0.039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.038 ± 0.042 vs 0.1254 (deviation 1.78σ)
Duration fit vs published	0.055 d vs 0.0746 d (deviation 0.43σ)
Mid-time O–C	0.6 min
Depth SNR	0.8
Residual scatter	19.72 %

Light curve

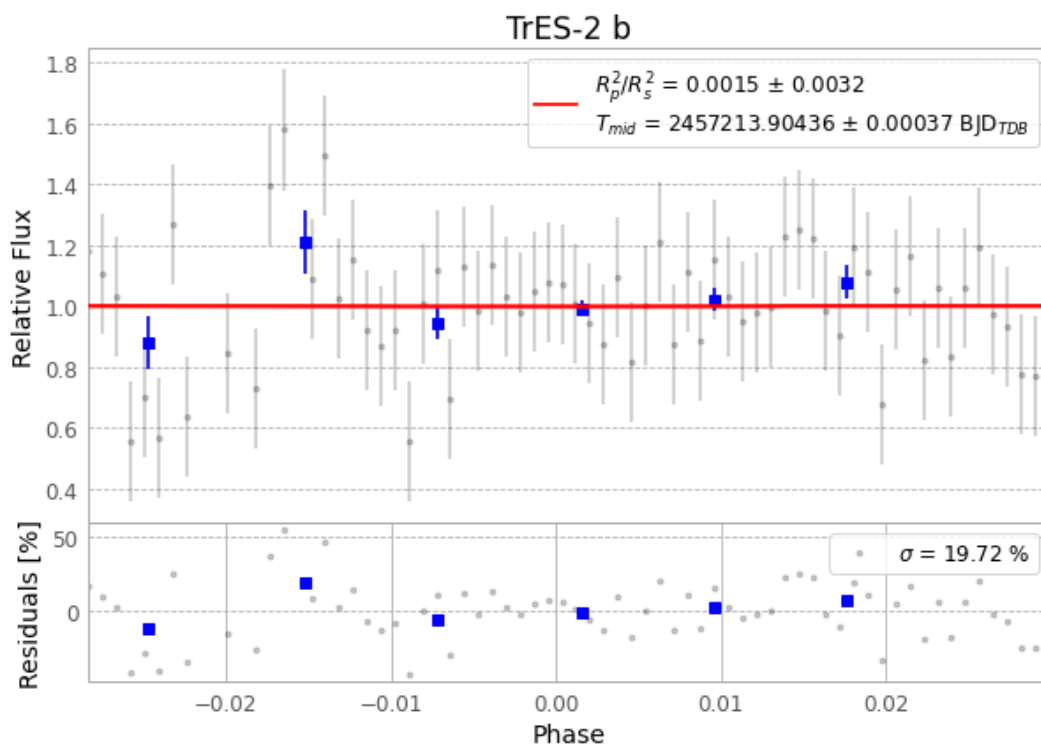


Figure 1 — FinalLightCurve_TrES-2 b_2015-07-10.png (SHA-256 b319c7e98294a0c9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [566, 143], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 10ae476ce894709f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), TRES-2150710075712.FITS → TRES-2150710112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-150710.log (75300f286711...), FinalLightCurve_TrES-2 b_2015-07-10.png (b319c7e98294...), FinalLightCurve_TrES-2 b_2015-07-10.csv (8ed43d7e6d59...)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 05:39Z.